```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

%matplotlib inline

df = pd.read_csv("bike_buyers-selected-columns.csv")

print("Dataset Shape:", df.shape)
display(df.head())

print(df.info())
```

```
Dataset Shape: (1000, 10)
```

	ID	Marital Status	Gender	Income	Children	Education	Occupation	Home Owner	Cars	Commute Distance
0	12496	Married	Female	40000.0	1.0	Bachelors	Skilled Manual	Yes	0.0	0-1 Miles
1	24107	Married	Male	30000.0	3.0	Partial College	Clerical	Yes	1.0	0-1 Miles
2	14177	Married	Male	80000.0	5.0	Partial College	Professional	No	2.0	2-5 Miles
3	24381	Single	NaN	70000.0	0.0	Bachelors	Professional	Yes	1.0	5-10 Miles
4	25597	Single	Male	30000.0	0.0	Bachelors	Clerical	No	0.0	0-1 Miles

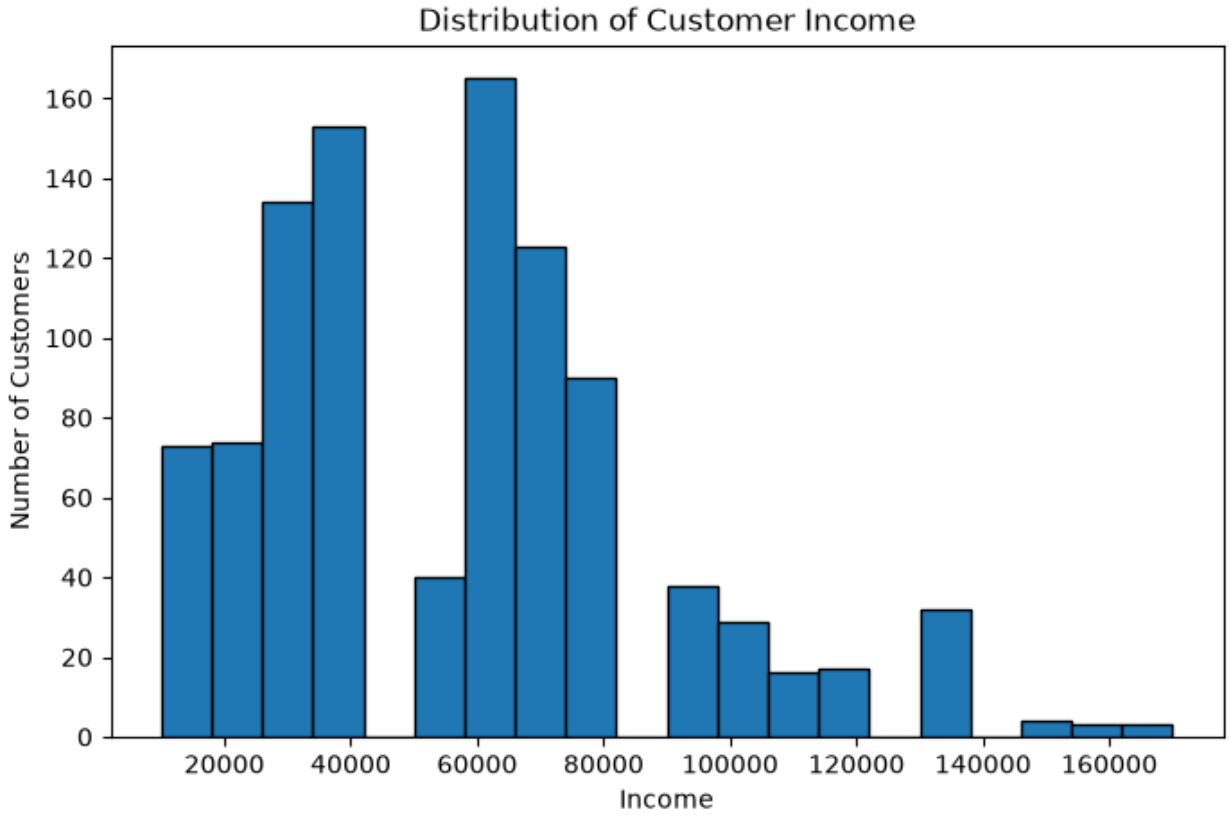
```
<class 'pandas.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 10 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   ID                1000 non-null   int64
 1   Marital Status    993 non-null    str
 2   Gender            989 non-null    str
 3   Income            994 non-null    float64
 4   Children          992 non-null    float64
 5   Education         1000 non-null   str
 6   Occupation        1000 non-null   str
 7   Home Owner        996 non-null    str
 8   Cars              991 non-null    float64
 9   Commute Distance  1000 non-null   str
dtypes: float64(3), int64(1), str(6)
memory usage: 78.3 KB
None
```

```python
plt.figure(figsize=(8, 5))
```

```
plt.hist(
    df['Income'].dropna(),
    bins=20,
    edgecolor='black'
)

plt.title('Distribution of Customer Income')
plt.xlabel('Income')
plt.ylabel('Number of Customers')

plt.show()
```

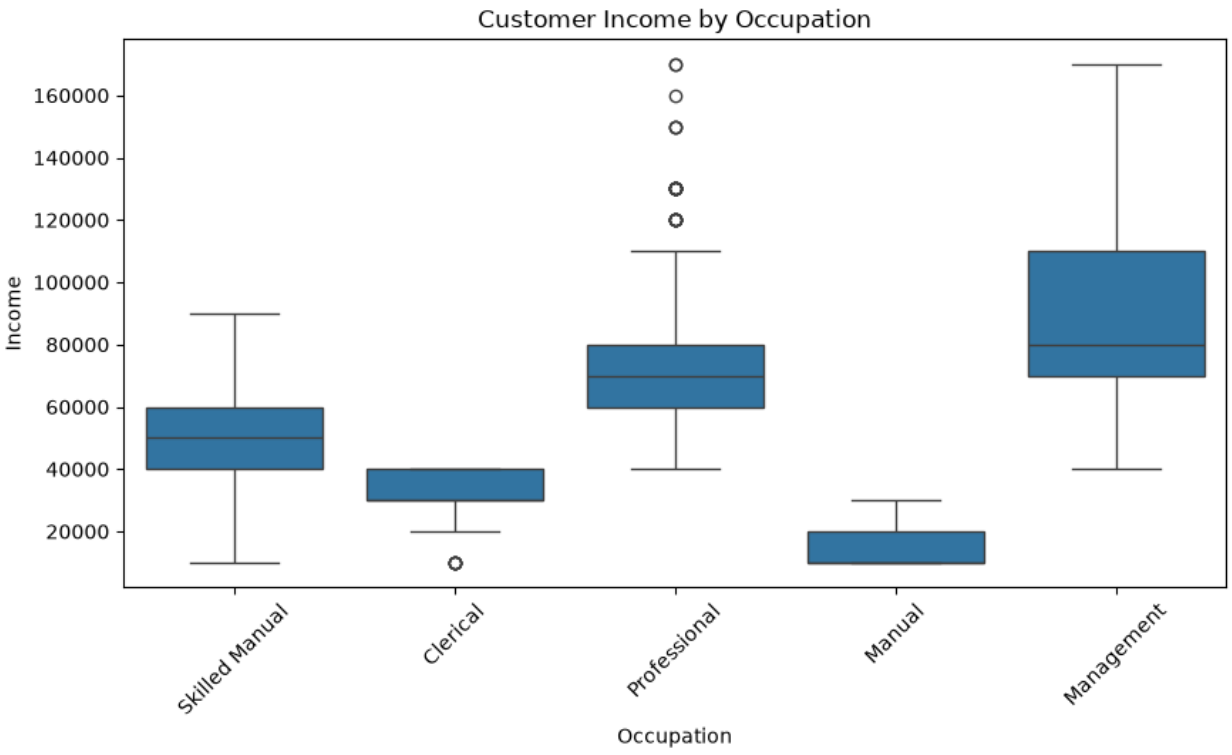



```
plt.figure(figsize=(10, 5))

sns.boxplot(
    data=df,
    x='Occupation',
    y='Income'
)

plt.title('Customer Income by Occupation')
plt.xlabel('Occupation')
plt.ylabel('Income')
```

```
plt.xticks(rotation=45)
plt.show()
```

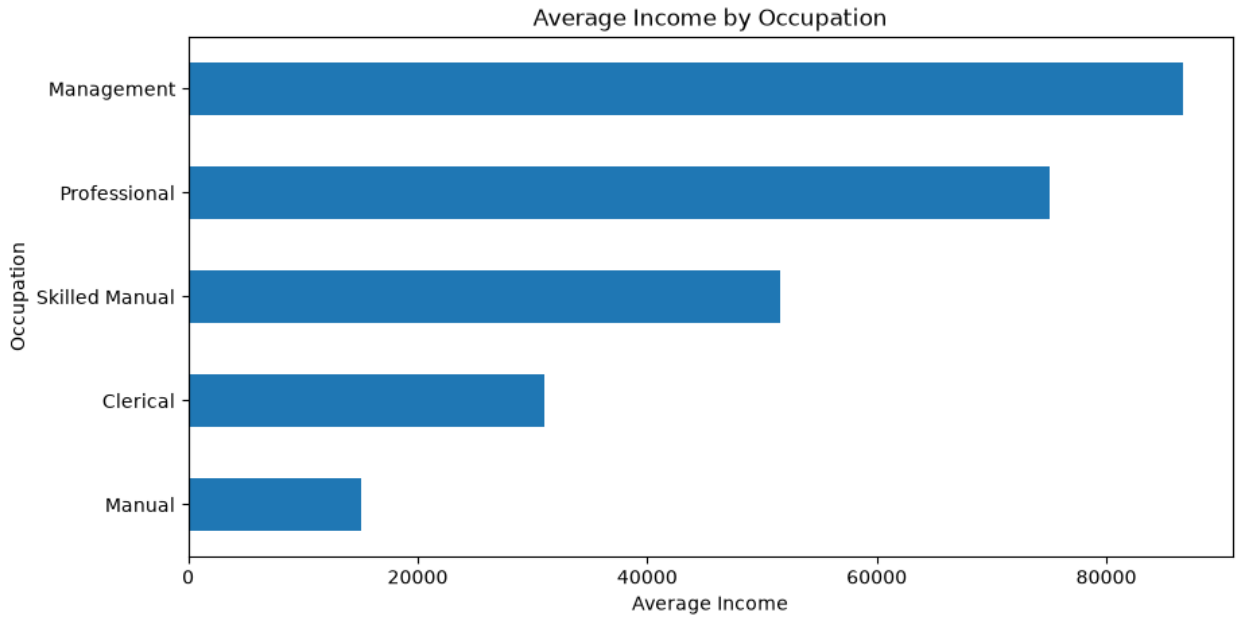



```
occupation_income = (
    df.groupby('Occupation')['Income']
    .mean()
    .sort_values(ascending=False)
    .sort_values()
)

plt.figure(figsize=(10, 5))
occupation_income.plot(kind='barh')

plt.title('Average Income by Occupation')
plt.xlabel('Average Income')
plt.ylabel('Occupation')

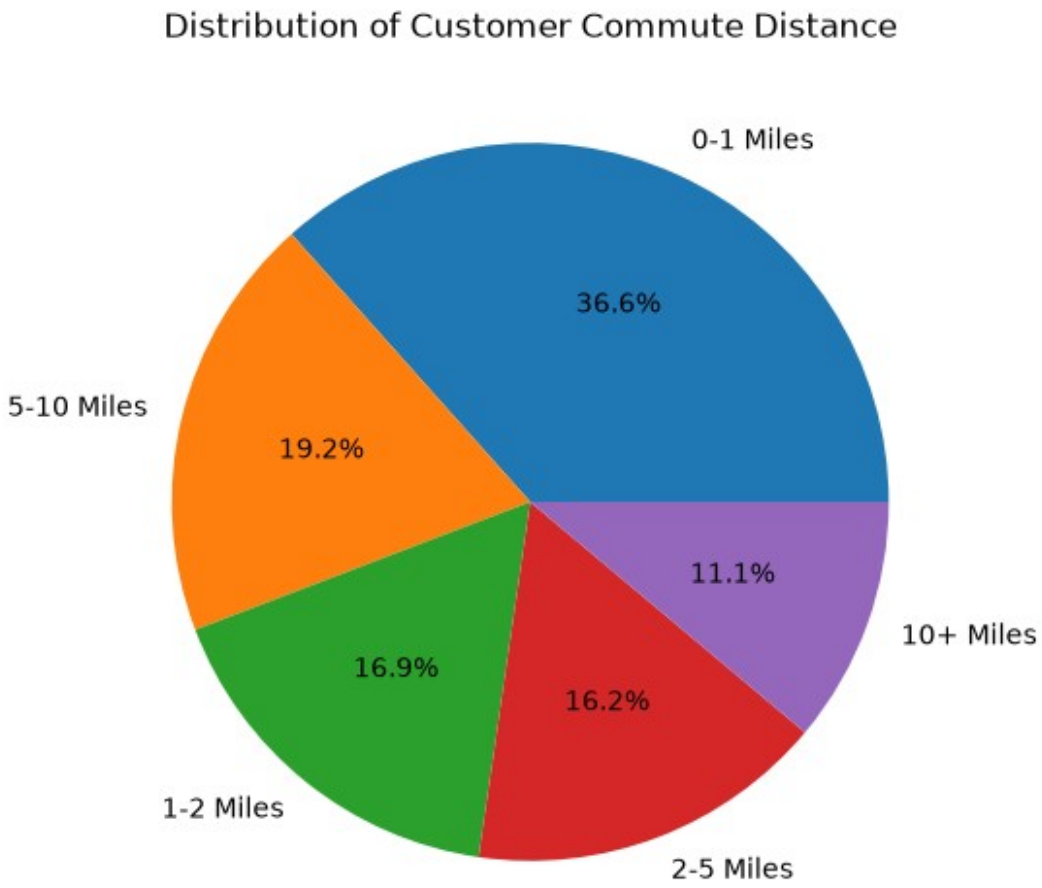
plt.show()
```




```
df['Commute Distance'].value_counts().plot(
    kind='pie',
    autopct='%1.1f%%',
    figsize=(8, 6)
)

plt.title('Distribution of Customer Commute Distance')
plt.ylabel('')

plt.show()
```

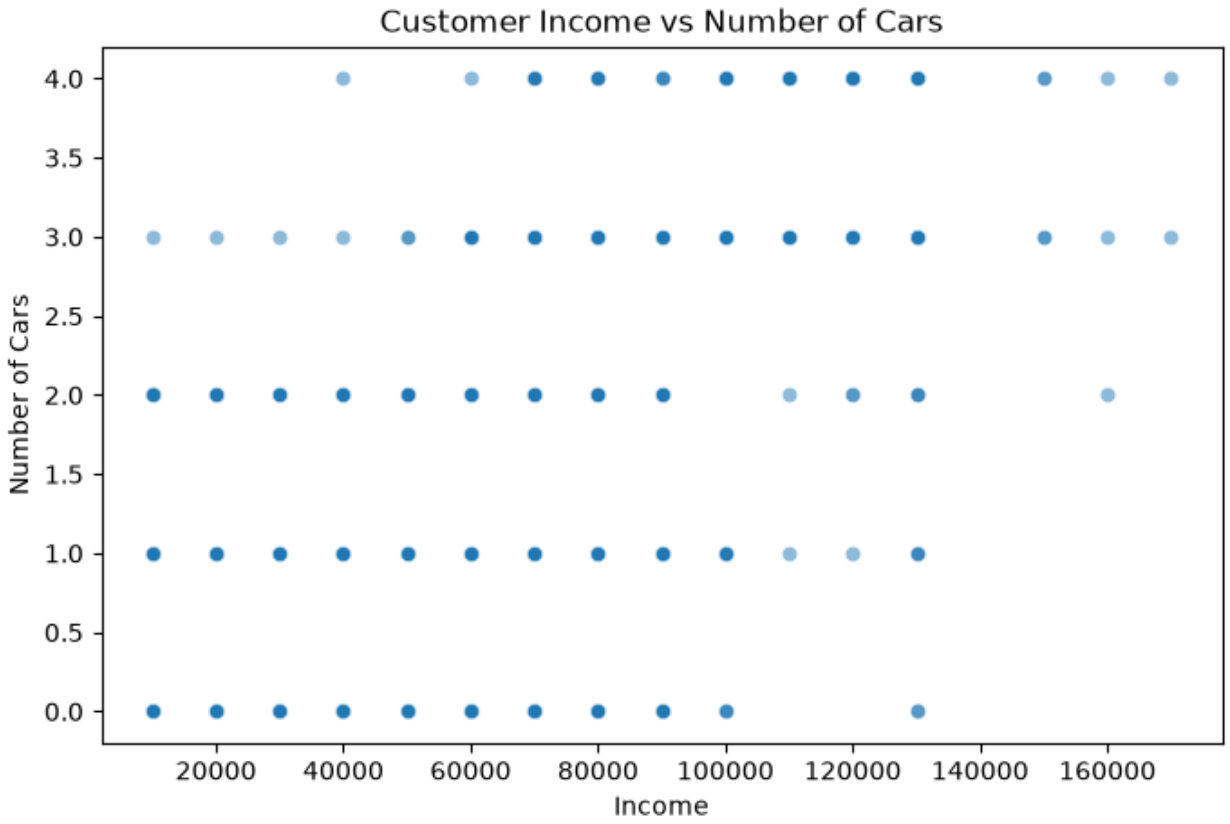



```
plt.figure(figsize=(8, 5))

sns.scatterplot(
    data=df,
    x='Income',
    y='Cars',
    alpha=0.5
)

plt.title('Customer Income vs Number of Cars')
plt.xlabel('Income')
plt.ylabel('Number of Cars')

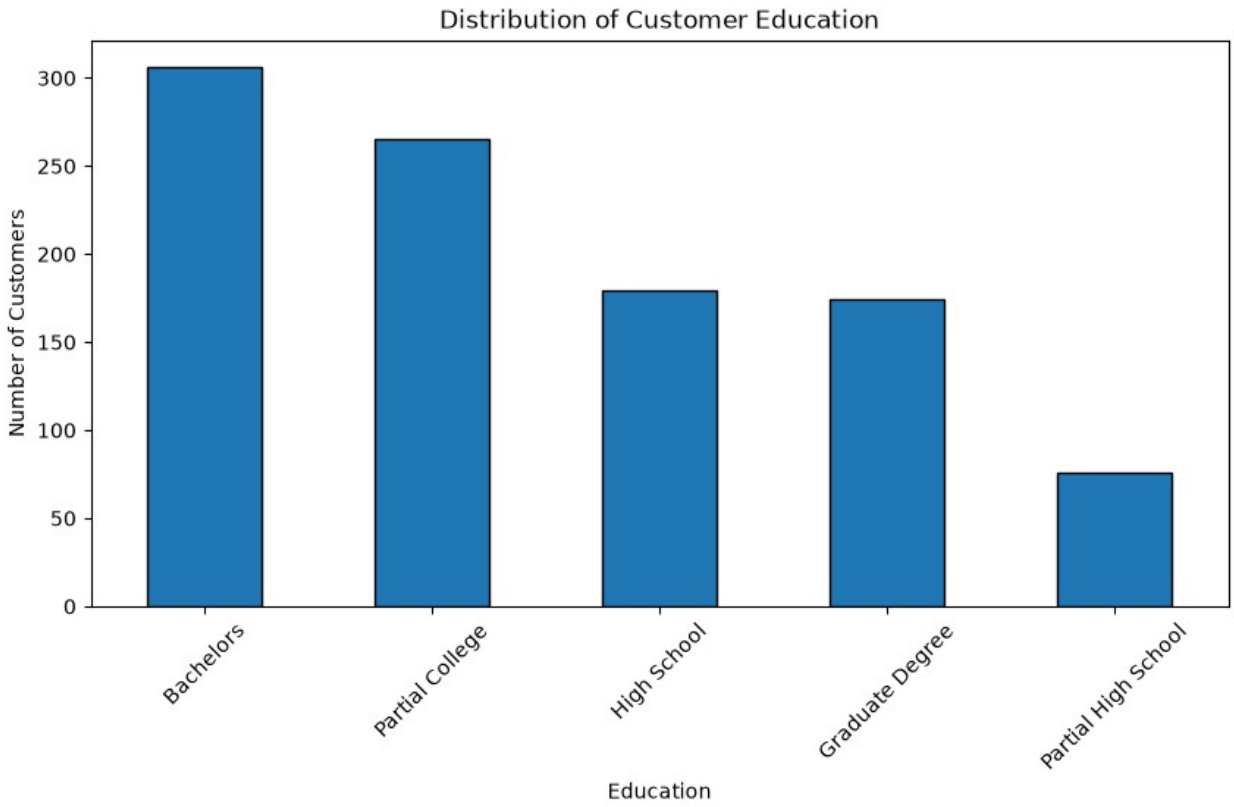
plt.show()
```




```
df['Education'].value_counts().plot(
    kind='bar',
    figsize=(10, 5),
    edgecolor='black'
)

plt.title('Distribution of Customer Education')
plt.xlabel('Education')
plt.ylabel('Number of Customers')

plt.xticks(rotation=45)
plt.show()
```

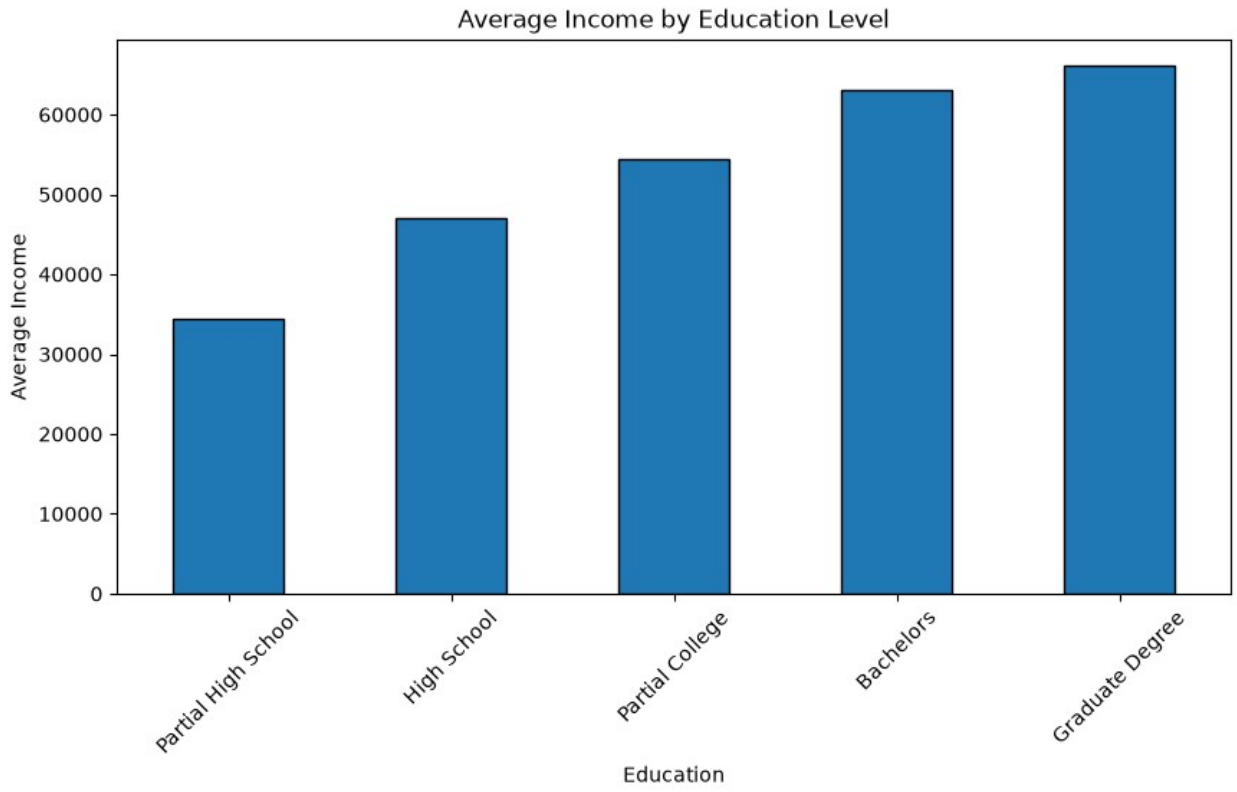



```
education_income = (
    df.groupby('Education')['Income']
    .mean()
    .sort_values()
)

plt.figure(figsize=(10, 5))
education_income.plot(kind='bar', edgecolor='black')

plt.title('Average Income by Education Level')
plt.xlabel('Education')
plt.ylabel('Average Income')

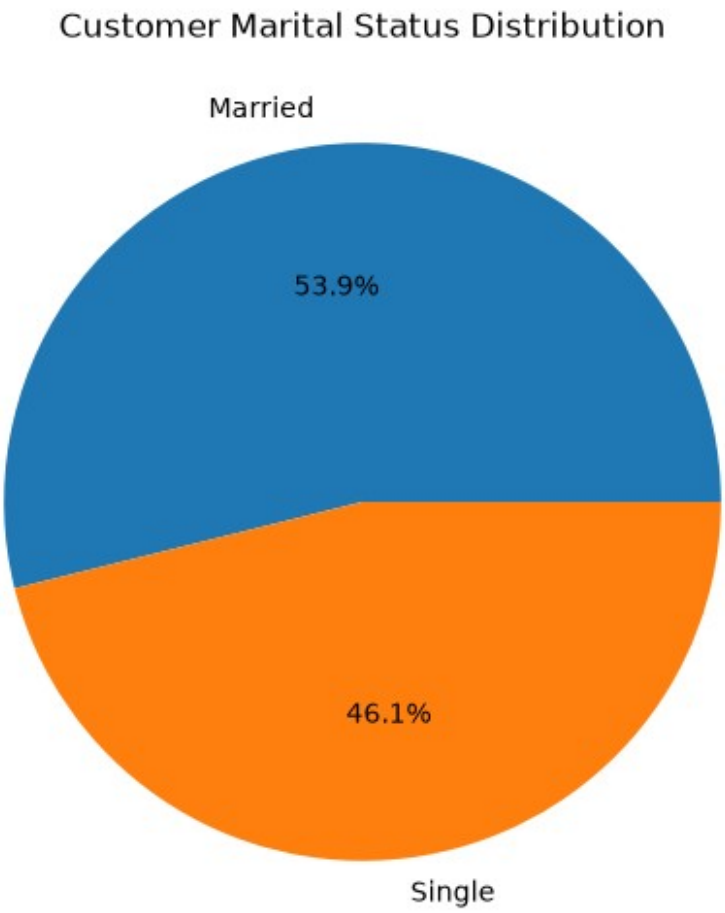
plt.xticks(rotation=45)
plt.show()
```




```
df['Marital Status'].value_counts().plot(
    kind='pie',
    autopct='%1.1f%%',
    figsize=(8, 6)
)

plt.title('Customer Marital Status Distribution')
plt.ylabel('')

plt.show()
```




```
print("Dataset Shape:", df.shape)

print("\nDataset Description:")
display(df.describe())

print("\nFirst 10 Records:")
display(df.head(10))
```

```
Dataset Shape: (1000, 10)

Dataset Description:
```

	ID	Income	Children	Cars
count	1000.000000	994.000000	992.000000	991.000000
mean	19965.992000	56267.605634	1.910282	1.455096
std	5347.333948	31067.817462	1.626910	1.121755
min	11000.000000	10000.000000	0.000000	0.000000
25%	15290.750000	30000.000000	0.000000	1.000000
50%	19744.000000	60000.000000	2.000000	1.000000
75%	24470.750000	70000.000000	3.000000	2.000000
max	29447.000000	170000.000000	5.000000	4.000000

First 10 Records:

```
      ID Marital Status  Gender    Income  Children
Education  \
0  12496        Married  Female   40000.0       1.0
Bachelors
1  24107        Married    Male   30000.0       3.0      Partial
College
2  14177        Married    Male   80000.0       5.0      Partial
College
3  24381         Single     NaN   70000.0       0.0
Bachelors
4  25597         Single    Male   30000.0       0.0
Bachelors
5  13507        Married  Female   10000.0       2.0      Partial
College
6  27974         Single    Male  160000.0       2.0          High
School
7  19364        Married    Male   40000.0       1.0
Bachelors
8  22155            NaN    Male   20000.0       2.0  Partial High
School
9  19280        Married    Male       NaN       2.0      Partial
College

       Occupation Home Owner  Cars Commute Distance
0  Skilled Manual        Yes   0.0        0-1 Miles
1        Clerical        Yes   1.0        0-1 Miles
2    Professional         No   2.0        2-5 Miles
3    Professional        Yes   1.0       5-10 Miles
4        Clerical         No   0.0        0-1 Miles
5          Manual        Yes   0.0        1-2 Miles
6      Management        NaN   4.0        0-1 Miles
7  Skilled Manual        Yes   0.0        0-1 Miles
8        Clerical        Yes   2.0       5-10 Miles
9          Manual        Yes   1.0        0-1 Miles
```